1 Lecture 01 (Aug 25) - Introduction to CS 374

1.1 Alphabets

Let Σ be an **alphabet**, an arbitrary **finite** set of **symbols** or **characters**

Examples include:

- $\Sigma = \{A, B, ..., Z\}$
- $\Sigma = \{0, 1\}$

1.2 Strings

A **string** or **word** is a **finite** sequence of symbols from an alphabet

Formally speaking, w is a string over Σ if it is one of

- the **empty string**
- an ordered pair (a, x), where $a \in \Sigma$ and x is a string over Σ

1.2.1 Example

$$\begin{aligned} STRING &= (S, (TRING)) \\ &= (S, (T, (RING))) \\ &= ... \\ &= (S, (T, (R, (I, (N, (G, \varepsilon)))))) \end{aligned}$$

1.3 Functions

Functions are defined recursively, where length $|w|$ represents the number of symbols in w

$$|w| := \begin{cases} 0 & \text{if } \ w = \varepsilon \\ 1 + |x| & \text{if } \ w = ax \end{cases}$$

1.3.1 Example

$$\begin{aligned} |HAT| &= 1 + |AT| \\ &= 1 + (1 + |T|) \\ &= 1 + (1 + (1 + |\varepsilon|)) \\ &= \boxed{3} \end{aligned}$$

1.4 Concatenation

Let **concatenation**, $w \bullet z$, of strings w and z be represented as

$$w \bullet z := \begin{cases} z & \text{if } w = \varepsilon \\ (a, x \bullet z) & \text{if } w = ax \end{cases}$$

1.4.1 Example

$$\begin{aligned} NOW \bullet HERE &= N \cdot (OW \bullet HERE) \\ &= N \cdot (O \cdot (W \bullet HERE)) \\ &= N \cdot (O \cdot (W \cdot (\varepsilon \bullet HERE))) \\ &= N \cdot (O \cdot (W \cdot HERE)) \\ &= \boxed{NOWHERE} \end{aligned}$$

1.5 Induction

- Consider an arbitrary object (probably of size n)
- **Explicitly** state a **strong** inductive hypothesis (i.e. `for all k < n`)
- Explicitly handle each case for the object
- Use inductive hypothesis when it's useful
- End by pointing out you handled all cases

1.5.1 Example 1

For every string w,

$$w \bullet \varepsilon = w$$

Let w be an arbitrary string. Assume for all strings x with $|x| < |w|$, that $x \bullet \varepsilon = x$

Suppose $w = \varepsilon$. Then,

$$\begin{aligned} w \bullet \varepsilon &= \varepsilon \bullet \varepsilon && \text{since } \ w = \varepsilon \\ &= \varepsilon && \text{by definition of concatenation } \ (\bullet) \\ &= w && \text{since } \ w = \varepsilon \end{aligned}$$

1.5.2 Example 2

Suppose $w = a \bullet x$ for symbol a and string x. Prove that in all cases, $w \bullet \varepsilon = w$

$$\begin{aligned} w \bullet \varepsilon &= (a \cdot x) \bullet \varepsilon && (w = ax) \\ &= a \cdot (x \bullet \varepsilon) && (\text{by definition of } \bullet) \\ &= a \cdot x && (\text{inductive hypothesis}) \\ &= w && (w = ax) \end{aligned}$$

Therefore, in all cases, $w \bullet \varepsilon = w$

1.5.3 Example 3

Let w and z be arbitrary, upon the assumption that for every string x such that $|x| < |w|$, that $|x \bullet z| = |x| + |z|$

1. Suppose that $w = \varepsilon$. Then,

$$\begin{aligned} |w \bullet z| &= |\varepsilon \bullet z| && w = \varepsilon \\ &= |z| && \text{by definition of concatenation } \bullet \\ &= 0 + |z| && \text{arithmetic} \\ &= |\varepsilon| + |z| && \text{by definition of length of string} \\ &= |w| + |z| && w = \varepsilon \end{aligned}$$

2. Suppose that $w = a \cdot x$ for some symbol a and string x. Then,

$$\begin{aligned} |w \bullet z| &= |(a \cdot x) \bullet z| && w = a \cdot x \\ &= |a \cdot (x \bullet z)| && \text{by definition of concatenation } \bullet \\ &= 1 + |x \bullet z| && \text{inductive hypothesis} \\ &= 1 + |x| + |z| && \text{by definition of length of string} \\ &= |w| + |z| && w = a \cdot x \end{aligned}$$